📐 Curve Sketching Guide

Brief Overview

This note covers **curve sketching** and was created from the [Calculus 1 Lecture 3.6: How to Sketch Graphs of Functions]() YouTube video (93 minutes). It walks you through finding **intercepts** and **derivatives**, spotting asymptotes, analyzing concavity, and assembling key points for a clear graph.

Key Points

- Identify x- and y-intercepts to anchor the graph.
- Use first and second derivatives to determine increasing/decreasing behavior and concavity.
- Locate asymptotes to understand vertical and horizontal behavior.
- Compile critical points and behavior tables to sketch the curve confidently.

📈 Curve Sketching: 7-Step Process

> **Curve sketching** uses **intercepts**, **extrema**, and **concavity** to draw an accurate graph without a calculator.

🧮 Step 1: Find Intercepts

- **X-intercepts**: Set $y = 0$ and solve for x
  - These are the **roots** of the function
  - Use **zero product property** if factored
- **Y-intercepts**: Set $x = 0$ and solve for y
  - Always exists unless function is undefined at $x = 0$

> *Only factor if it's **easy** (quadratic or lower, or already factored)*

Example: For $f(x) = (x + 2)(x - 1)^2$

- X-intercepts: $x = -2$ and $x = 1$
- Y-intercept: $f(0) = 2$

🔍 Step 2: Asymptotes (Rational Functions Only)

- **Vertical asymptotes**: Set **denominator = 0**
  - If factor can be **crossed out** → **hole**
  - If factor **cannot be crossed out** → **vertical asymptote**
- **Horizontal asymptotes**: Take $\lim_{x \to \pm\infty} f(x)$
  - Divide by **highest power in denominator**

📊 Step 3: First Derivative Test

- Find $f'(x)$ and set $f'(x) = 0$ → **critical numbers**
- Use sign chart to determine:
  - **Increasing** where $f'(x) > 0$
  - **Decreasing** where $f'(x) < 0$
  - **Relative max/min** at critical numbers

📉 Step 4: Second Derivative Test

- Find $f''(x)$ and set $f''(x) = 0$ → **possible inflection points**
- Use sign chart to determine:
  - **Concave up** where $f''(x) > 0$
  - **Concave down** where $f''(x) < 0$
  - **Inflection point** where concavity changes

📋 Step 5: Create Summary Table

Interval	$f'(x)$	$f''(x)$	Behavior
$(-\infty, -1)$	+	-	Increasing, concave down
$(-1, 0)$	-	-	Decreasing, concave down
$(0, 1)$	-	+	Decreasing, concave up
$(1, \infty)$	+	+	Increasing, concave up

🎯 Step 6: Find Key Points

- **X-intercepts**: $(-2, 0)$ and $(1, 0)$
- **Y-intercept**: $(0, 2)$
- **Relative max**: $(-1, 4)$
- **Relative min**: $(1, 0)$
- **Inflection point**: $(0, 2)$

> *Label **IP** for inflection points and use **dashed lines** at extrema to show graph boundaries*

✏️ Step 7: Sketch the Graph

1. Plot all **key points**
2. Draw **asymptotes** (if any)
3. Follow **table behavior** from left to right:
    - **Increasing concave down** → through $(-2, 0)$ to **relative max** at $(-1, 4)$
    - **Decreasing concave down** → to **inflection point** at $(0, 2)$
    - **Decreasing concave up** → to **relative min** at $(1, 0)$
    - **Increasing concave up** → indefinitely

🧠 Rational Function Example: $f(x) = \frac{x^2-1}{x^3}$

Step 1: Intercepts

- **X-intercepts**: Set numerator = 0 → $x^2 - 1 = 0$ → $x = \pm 1$
- **Y-intercept**: $f(0)$ is **undefined** → **no Y-intercept**

Step 2: Asymptotes

- **Vertical asymptote**: Set denominator = 0 → $x^3 = 0$ → $x = 0$
- **Horizontal asymptote**:
    - $\lim_{x \to \pm\infty} \frac{x^2-1}{x^3} = \lim_{x \to \pm\infty} \frac{1}{x} - \frac{1}{x^3} = 0$
    - **Horizontal asymptote at** $y = 0$

Steps 3-7: Continue with derivative tests and sketch as before

🧮 Advanced Rational Function Example: $f(x) = \frac{x^2-8}{x^2-16}$

🔍 Step 1: Intercepts

- **X-intercepts**: Set numerator = 0 → $x^2 - 8 = 0$ → $x = \pm\sqrt{8} = \pm 2\sqrt{2}$
- **Y-intercept**: $f(0) = \frac{-8}{-16} = \frac{1}{2}$

📊 Step 2: Asymptotes

- **Vertical asymptotes**: Set denominator = 0 → $x^2 - 16 = 0$ → $x = \pm 4$
  - These cannot be simplified with numerator → **true asymptotes**
- **Horizontal asymptote**: $\lim_{x\to\pm\infty} \frac{x^2-8}{x^2-16} = \frac{1}{1} = 1$
  - Since degrees are equal, divide leading coefficients

📈 Step 3: First Derivative

> Use quotient rule: $\frac{low \cdot high' - high \cdot low'}{low^2}$

$$f'(x) = \frac{(x^2-16)(2x)-(x^2-8)(2x)}{(x^2-16)^2} = \frac{2x(x^2-16-x^2+8)}{(x^2-16)^2} = \frac{-16x}{(x^2-16)^2}$$

- **Critical numbers**: Set numerator = 0 → $x = 0$
- Set denominator = 0 → $x = \pm 4$ (asymptotes affect increasing/decreasing behavior)

📉 Step 4: Second Derivative

$$f''(x) = \frac{48(x^2+16)}{(x^2-16)^3}$$

- **Possible inflection points**: Set numerator = 0 → $x^2 + 16 = 0$ → **no real solutions**
- Set denominator = 0 → $x = \pm 4$ (asymptotes affect concavity)

📋 Step 5: Test Intervals

Interval	Test Point	$f'(x)$ Sign	$f''(x)$ Sign	Behavior
$(-\infty, -4)$	-5	+	+	**Increasing**, **concave up**
$(-4, 0)$	-1	+	-	**Increasing**, **concave down**

$(0, 4)$	1	-	-	**Decreasing, concave down**
$(4, \infty)$	5	-	+	**Decreasing, concave up**

🎯 Step 6: Key Points

- **X-intercepts**: $(-2\sqrt{2}, 0)$ and $(2\sqrt{2}, 0)$
- **Y-intercept**: $(0, \frac{1}{2})$
- **Relative maximum**: $(0, \frac{1}{2})$
- **No inflection points** (concavity changes only at asymptotes)

✏️ Step 7: Sketch the Graph

1. Draw **horizontal asymptote** at $y = 1$
2. Draw **vertical asymptotes** at $x = \pm 4$
3. Plot **intercepts** at $x = \pm 2\sqrt{2} \approx \pm 2.8$
4. Use **test results**:
    - Left side: **increasing** from -∞, approaching $y = 1$
    - Between $-\infty$ to -4: **increasing concave up**
    - Between -4 to 0: **increasing concave down** to **maximum** at $(0, \frac{1}{2})$
    - Between 0 to 4: **decreasing concave down**
    - From 4 to ∞: **decreasing concave up**, approaching $y = 1$

🧠 Key Insights

> **Critical numbers** include both where **derivative = 0** AND where **derivative is undefined**

- At **asymptotes**, function may change **increasing/decreasing** behavior
- **Concavity** changes can occur at **asymptotes** even if no inflection points exist
- When **degrees equal** in rational functions, **horizontal asymptote** = ratio of leading coefficients

🔍 Completing the Second Derivative

> **Second derivative** reveals **concavity** - how the slope is changing (the curvature of the graph)

📊 Concavity Types

- **Increasing concave up**: Graph curves upward while rising
- **Increasing concave down**: Graph curves downward while rising
- **Decreasing concave up**: Graph curves upward while falling
- **Decreasing concave down**: Graph curves downward while falling

🧮 Finding Possible Inflection Points

- **Set numerator = 0**: Gives possible inflection points (actual values to plug into function)
- **Set denominator = 0**: Shows where second derivative is undefined (likely at asymptotes)
- For $f(x) = \frac{x^2-8}{x^2-16}$:
  - **Numerator**: $3x^2 + 16 = 0$ → $x^2 = -\frac{16}{3}$ → **No real solutions**
  - **Denominator**: $(x^2 - 16)^3 = 0$ → $x = \pm 4$ (same asymptotes)

📋 Step 5: Test Intervals Analysis

Interval	Test Point	$f'(x)$ Sign	$f''(x)$ Sign	Behavior
$(-\infty, -4)$	-5	+	+	Increasing, concave up
$(-4, 0)$	-1	+	-	Increasing, concave down
$(0, 4)$	1	-	-	Decreasing, concave down
$(4, \infty)$	5	-	+	Decreasing, concave up

📍 Step 6: Key Points Summary

Available Points to Plot

- **X-intercepts**: $(-2\sqrt{2}, 0)$ and $(2\sqrt{2}, 0) \approx (\pm 2.8, 0)$
- **Y-intercept**: $(0, \frac{1}{2})$
- **Relative maximum**: $(0, \frac{1}{2})$ (only extremum)

Points NOT Available

- **At asymptotes $x = \pm 4$**: Function is **undefined**
- **No inflection points**: Concavity changes only at asymptotes

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✏️ Step 7: Graph Construction Strategy

Using Asymptotes + First Derivative Only

> **When asymptotes exist**, you can often sketch the graph using just **asymptotes** and **increasing/decreasing** behavior

Interval $(-\infty, -4)$

- **Increasing** with **horizontal asymptote** at $y = 1$
- **Vertical asymptote** at $x = -4$
- **Only possibility**: Curve approaches $y = 1$ from below, rises to ∞ at $x = -4$

Interval $(-4, 0)$

- **Increasing** then **decreasing** at $x = 0$
- **Relative maximum** at $(0, \frac{1}{2})$
- **Only possibility**: Comes from $-\infty$ at $x = -4$, rises to max at $(0, \frac{1}{2})$

Interval $(0, 4)$

- **Decreasing** with **vertical asymptote** at $x = 4$
- **Only possibility**: Falls from max at $(0, \frac{1}{2})$ to $-\infty$ at $x = 4$

Interval $(4, \infty)$

- **Decreasing** with **horizontal asymptote** at $y = 1$
- **Only possibility**: Comes from ∞ at $x = 4$, falls toward $y = 1$

Concavity Confirmation

- Each interval's concavity matches the **only possible** curve shape

- **No relative minimum** exists (confirmed by table analysis)
- **Second derivative** validates the **first derivative** sketch

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🎯 Critical Insights for Rational Functions

> **When degrees are equal in numerator/denominator**, **horizontal asymptote** = ratio of leading coefficients

Asymptote Behavior Rules

- **Vertical asymptotes**: Function approaches $\pm\infty$
- **Horizontal asymptotes**: Function approaches finite value as $x \to \pm\infty$
- **Can cross horizontal asymptotes** (unlike vertical asymptotes)

Graphing Shortcuts

1. **Plot asymptotes** first
2. **Plot intercepts** and extrema
3. **Use increasing/decreasing** to connect points
4. **Verify with concavity** if needed
5. **No relative minimum** in this function - only **relative maximum** at $(0, \frac{1}{2})$